AI 101 FOR MECHANICAL ENGINEERS

MODULE 15 / LESSON 1

Uncertainty and changing conditions



Undergraduate course

Lesson outcomes

Distinguish aleatoric and epistemic uncertainty.

Explain calibration, operating coverage and distribution shift.

Aleatoric and epistemic uncertainty

Aleatoric uncertainty refers to variation in observations or unresolved processes, such as measurement noise.

Epistemic uncertainty comes from limited knowledge, data or model assumptions.

More informative observations may reduce some epistemic uncertainty, but an inadequate model can remain wrong.

Calibration and extrapolation

Calibration examines whether stated probabilities or intervals agree with observed frequencies under appropriate conditions. Model confidence is not automatically calibrated.

Interpolation stays within covered conditions. Extrapolation reaches beyond them. Being inside simple input bounds still does not guarantee representative coverage.

Distribution shift and drift

Distribution shift means the deployment population differs from development. Drift describes change over time, such as wear, a new control regime or sensor recalibration.

Monitor inputs and outcomes and define when to defer or retrain. A seed cannot remove shift.

Lesson vocabulary 1

Aleatoric uncertainty: Variation inherent in observations or unresolved processes. Example: Measurement noise in a temperature reading.

Epistemic uncertainty: Uncertainty due to limited knowledge, data or assumptions. Example: Sparse measurements in a load regime.

Calibration: Agreement of stated probabilities or intervals with observed frequencies under defined conditions. Example: Check fault probability groups on independent data.

Lesson vocabulary 2

Interpolation: Estimating within covered input conditions. Example: Predict within a sampled operating region.

Extrapolation: Estimating beyond covered input conditions. Example: Use a model at a higher temperature than observed.

Distribution shift: A difference between development and deployment data distributions. Example: Transfer from one plant type to another.

Lesson vocabulary 3

Drift: Change in data or relationships over time. Example: Sensor wear alters readings.

Worked exercise

Can a high model confidence justify using it outside the validated operating domain?

Pause and write an answer before moving to the explanation.

Exercise explanation

No. Confidence can be poorly calibrated, especially under distribution shift. Conditions outside the evaluated domain require new evidence or deferral.

Concept reference table 1

Term	Meaning	ME example
Aleatoric uncertainty	Unresolved observed variation	Measurement noise
Epistemic uncertainty	Limited knowledge or evidence	Sparse load data
Calibration	Confidence versus observed frequency	Reported probabilities

Teaching example. Read the table alongside the worked explanation.

Concept reference table 2

Term	Meaning	ME example
Drift	Change over time	Ageing sensor

Teaching example. Read the table alongside the worked explanation.

Knowledge check 1

Measurement noise is usually an example of…

A. Aleatoric uncertainty

B. A perfect ground truth

C. An epoch

D. An API

Knowledge check 2

Sparse data in an operating regime contributes to…

A. Epistemic uncertainty

B. Guaranteed accuracy

C. No uncertainty

D. A fixed unit conversion

Knowledge check 3

Sensor ageing that changes inputs over time is…

A. Drift

B. A token

C. A hidden layer

D. A class label by definition

Knowledge check 4

Being inside the input bounds guarantees trustworthy interpolation.

A. True

B. False

Knowledge-check explanations

Check 1: A. It is variation in observations.

Check 2: A. Limited knowledge supports epistemic uncertainty.

Knowledge-check explanations

Check 3: A. Drift describes change over time.

Check 4: B. Coverage, model assumptions and validation still matter.

Lesson recap

Distinguish aleatoric and epistemic uncertainty.

Explain calibration, operating coverage and distribution shift.

Apply the ideas to this question: Can a high model confidence justify using it outside the validated operating domain?

Sources and further reading

Primary source: AI 101 for Mechanical Engineers, Module 15, Lesson 1: Uncertainty and changing conditions.

Worked cases and numerical examples are teaching material. Synthetic evidence does not validate real equipment.

Google Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

scikit-learn User Guide
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